```
        image.exit()
        .remove();

        image.enter().append("image")
        .attr("xlink:href", function(d) { return "http://" +
            ["a", "b", "c", "d"][Math.random() * 4 | 0] +
            ".tiles.mapbox.com/v3/examples.map-zgrqqx0w/" + d[2] +
            "/" + d[0] + "/" + d[1] + ".png"; })
        .attr("width", 1)
        .attr("height", 1)
        .attr("x", function(d) { return d[0]; })
        .attr("y", function(d) { return d[1]; });

        projection
        .scale(zoom.scale() / 2 / Math.PI)
        .translate(zoom.translate());

        d3.selectAll("path.countries")
        .attr("d", path);

               d3.selectAll("circle").attr("cx", function(d) {
                           return projection([d.x,d.y])[0];
                       })
                     .attr("cy", function(d) {
                         return projection([d.x,d.y])[1];
                     });
    };
};
```

Note that we're not taking zoom.scale() directly like we did before, but processing it to get the properly formatted scale for our Mercator projection.

7.6 Further reading for web mapping

As I said in the beginning of this chapter, the things you can do with D3's mapping capabilities would fill an entire book. Following are a few other capabilities we didn't cover in this chapter.

7.6.1 Transform zoom

The method we used for our zoom behavior in this chapter is known as *projection zoom* and recalculates mathematically the shape of features based on a change in scale and translation. But if you're using a projection that's flat like Mercator, then you can achieve faster performance by tying the change in scale and translate of the zoom behavior to your features' SVG transform. One issue you'll run into is that font size and stroke width are affected by SVG transform, and so you'll need to adjust those settings on the fly.

7.6.2 Canvas drawing

The `.context` function `d3.geo.path` allows you to easily draw your vector data to a `<canvas>` element, which can dramatically improve speed in certain cases. It also allows you to use `.toDataURL()` to dynamically create a PNG for users to save or share on social media.